Legal aspects of IT security
Master in cybersecurity

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AI and cybersecurity

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Summary

I. The AI system
II. A « risk-based » approach
III. Cybersecurity obligations for AI systems
IV. Focus on remote biometric identification systems
V. To go further

-> AI Act of 13 June 2024

I. The AI system

AI system = “ *a machine-based system that is designed to operate with varying levels of autonomy and that may exhibit adaptiveness after deployment, and that, for explicit or implicit objectives, infers, from the input it receives, how to generate outputs such as predictions, content, recommendations, or decisions that can influence physical or virtual environments*” (AI Act, art. 3, 1°)

II. A « risk-based approach »

A risk = “ *the combination of the probability of an occurrence of harm and the severity of that harm*” (AI Act, art. 3, 2°)

Risk-based approach = type and content of rules // intensity and scope of the risks that AI systems can generate

-> 3 levels of AI

II. A « risk-based approach »

-> 3 levels of AI :

- Prohibited AI practices
- High-risk AI Systems
- Minimal or Minor-Risk AI Systems

II. A « risk-based approach »

- **Prohibited AI practices** (AI Act, art. 5)

- Deploying subliminal, manipulative, or deceptive techniques to distort behavior and impair informed decision-making, causing significant harm
- Exploiting vulnerabilities related to age, disability, or socio-economic circumstances to distort behavior, causing significant harm
- Biometric categorization systems that infer sensitive attributes like race, political opinions, trade union membership, religious or philosophical beliefs, sex life, or sexual orientation (with limited exceptions for labeling or filtering lawfully acquired datasets and for law enforcement use)
- Social scoring systems, such as systems that evaluate or classify individuals or groups based on social behavior or personal traits, causing them harm
- Assessing the risk of an individual committing criminal offenses based solely on profiling or personality traits (with limited exceptions)
- Compiling facial recognition databases through untargeted scraping of facial images from the internet or CCTV footage
- Inferring emotions in workplaces or educational institutions
- Real-time remote biometric identification in public places for law enforcement (with certain exceptions).

II. A « risk-based approach »

- **High-risk AI Systems** (AI Act, art. 6)

2 categories :

- AI systems where both of the following conditions are fulfilled :
  - the AI system is intended to be used as a safety component of a product, or the AI system is itself a product, covered by the Union harmonisation legislation listed in Annex I*
  - the product whose safety component pursuant to point (a) is the AI system, or the AI system itself as a product, is required to undergo a third-party conformity assessment, with a view to the placing on the market or the putting into service of that product pursuant to the Union harmonisation legislation listed in Annex I.
- AI systems referred to in Annex III*

II. A « risk-based approach »

- **High-risk AI Systems** (AI Act, art. 6)

Harmonisation legislation from Annex I*

- Machinery
- Safety of toys
- Recreational craft and personal watercraft
- Lifts and safety components for lifts
- Equipment and protective systems intended for use in potentially explosive atmospheres
- The making available on the market of radio equipment
- The making available on the market of pressure equipment
- Cableway installation
- Personal protective equipment
- […]

II. A « risk-based approach »

- **High-risk AI Systems** (AI Act, art. 6)

Harmonisation legislation from Annex I*

- […]
- Appliances burning gaseous fuels
- Medical devices
- In vitro diagnostic medical devices
- The field of civil aviation security
- The approval and market surveillance of two- or three-wheel vehicles and quadricycles
- The approval and market surveillance of agricultural and forestry vehicles
- Marine equipment
- The interoperability of the rail system within the European Union
- The approval and market surveillance of motor vehicles and their trailers, and of systems, components and separate technical units intended for such vehicles
- Type-approval requirements for motor vehicles and their trailers, and systems, components and separate technical units intended for such vehicles, as regards their general safety and the protection of vehicle occupants and vulnerable road users
- The field of civil aviation and establishing a European Union Aviation Safety Agency

II. A « risk-based approach »

- **High-risk AI Systems** (AI Act, art. 6)

AI systems referred to in Annex III* :

Some AI systems in the following areas :

- Biometrics
- Critical infrastructure
- Education and vocational training
- Employment, workers' management and access to self-employment
- Access to and enjoyment of essential private services and essential public services and benefits
- Law enforcement, in so far as their use is permitted under relevant Union or national law
- Migration, asylum and border control management, in so far as their use is permitted under relevant Union or national law
- Administration of justice and democratic processes

II. A « risk-based approach »

- **Minimal or Minor-Risk AI Systems**

-> No unacceptable or high-risks

Limited requirements : only transparency obligations

III. Cybersecurity obligations for AI systems

Distinction :

- High-risk systems
- Minimal or Minor-Risk AI Systems

III. Cybersecurity obligations for AI systems

- **High-risk systems**

"*High-risk AI systems shall be designed and developed in such a way that they achieve an <u>appropriate level</u> of accuracy, robustness, and cybersecurity, and that they perform consistently in those respects throughout their lifecycle*" (AI Act, art. 15, §1)

-> Appropriate technical solutions to the relevant circumstances and risks, that include : measures to prevent, detect, respond to, resolve and control for attacks trying to manipulate the training data set (data poisoning), or pre-trained components used in training (model poisoning), inputs designed to cause the AI model to make a mistake (adversarial examples or model evasion), confidentiality attacks or model flaws

III. Cybersecurity obligations for AI systems

- **Minimal or Minor-Risk AI Systems**

Obligation for providers of general-purpose AI models with systemic risk to "*ensure an <u>adequate level</u> of cybersecurity protection for the general-purpose AI model with systemic risk and the physical infrastructure of the model*" (AI Act, art. 55, §1, d))

IV. Focus on remote biometric identification systems

- biometric identification = "*the automated recognition of physical, physiological, behavioural, or psychological human features for the purpose of establishing the identity of a natural person by comparing biometric data of that individual to biometric data of individuals stored in a database*" (AI Act, art. 3, 35°)
- remote biometric identification system* = "*an AI system for the purpose of identifying natural persons, without their active involvement, typically at a distance through the comparison of a person's biometric data with the biometric data contained in a reference database*" (AI Act, art. 3, 41°)

IV. Focus on remote biometric identification systems

- **remote biometric identification system***

Distinction :

- real-time remote biometric identification system = "*a remote biometric identification system, whereby the capturing of biometric data, the comparison and the identification all occur without a significant delay, comprising not only instant identification, but also limited short delays in order to avoid circumvention*"(AI Act, art. 3, 42°)
- post-remote biometric identification system = "*a remote biometric identification system other than a real-time remote biometric identification system*" (AI Act, art. 3, 43°)

IV. Focus on remote biometric identification systems

What are the obligations for remote biometric identification systems?

Distinction :

- Real-time remote biometric identification system for the purposes of law enforcement
- Post-remote biometric identification system in the framework of an investigation for the targeted search of a person suspected or convicted of having committed a criminal offence

IV. Focus on remote biometric identification systems

- **Real-time remote biometric identification system for the purposes of law enforcement**

-> Prohibited AI practice EXCEPT if it is strictly necessary for one of the following purposes :

- the targeted search for specific victims of abduction, trafficking in human beings or sexual exploitation of human beings, as well as the search for missing persons;
- the prevention of a specific, substantial and imminent threat to the life or physical safety of natural persons or a genuine and present or genuine and foreseeable threat of a terrorist attack;
- the localisation or identification of a person suspected of having committed a criminal offence, for the purpose of conducting a criminal investigation or prosecution or executing a criminal penalty for offences referred to in Annex II* and punishable in the Member State concerned by a custodial sentence or a detention order for a maximum period of at least four years

IV. Focus on remote biometric identification systems

- **Real-time remote biometric identification system for the purposes of law enforcement**

Annex II* :

— terrorism,

— trafficking in human beings,

— sexual exploitation of children, and child pornography,

— illicit trafficking in narcotic drugs or psychotropic substances,

— illicit trafficking in weapons, munitions or explosives,

— murder, grievous bodily injury,

— illicit trade in human organs or tissue,

— illicit trafficking in nuclear or radioactive materials,

— kidnapping, illegal restraint or hostage-taking,

— crimes within the jurisdiction of the International Criminal Court,

— unlawful seizure of aircraft or ships,

— rape,

— environmental crime,

— organised or armed robbery,

— sabotage,

— participation in a criminal organisation involved in one or more of the offences listed above

IV. Focus on remote biometric identification systems

- **Real-time remote biometric identification system for the purposes of law enforcement**

Conditions of application :

- only to confirm the identity of the specifically targeted individual, and it shall take into account the following elements:

(a) the nature of the situation giving rise to the possible use, in particular the seriousness, probability and scale of the harm that would be caused if the system were not used;

(b) the consequences of the use of the system for the rights and freedoms of all persons concerned, in particular the seriousness, probability and scale of those consequences

- necessary and proportionate safeguards and conditions (fundamental rights impact assessment)
- a prior authorisation granted by a judicial authority or an independent administrative authority
- temporal, geographic and personal limitations
- Notification to the relevant market surveillance authority and the national data protection authority

IV. Focus on remote biometric identification systems

- **Post-remote biometric identification system in the framework of an investigation for the targeted search of a person suspected or convicted of having committed a criminal offence**

High risk AI system, EXCEPT if it is used for the sole purpose of which is to confirm that a specific natural person is the person he or she claims to be

IV. Focus on remote biometric identification systems

- **Post-remote biometric identification system in the framework of an investigation for the targeted search of a person suspected or convicted of having committed a criminal offence**

Conditions of application :

- The deployer shall request an authorisation, ex ante, or without undue delay and no later than 48 hours, by a judicial authority or an administrative authority whose decision is binding and subject to judicial review, for the

use of that system

EXCEPT when it is used for the initial identification of a potential suspect based on objective and verifiable

facts directly linked to the offence

- Limitation to what is strictly necessary for the investigation of a specific

criminal offence

- Annual reports to the relevant market surveillance and national data protection authorities

V. To go further

CRIDS serie « 11 shades of AI Act »

https://www.crids.eu/research/issues/artificial-intelligence-big-data-iot

Any questions?

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Thanks for your attention!

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